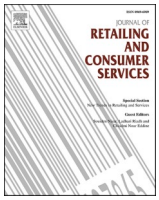
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Mere copycat? The effects of human versus human-like virtual influencers on brand endorsement effectiveness: A moderated serial-mediation model

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ABSTRACT

The emergence of virtual influencers, who possess realistic human-like appearances and intelligence, has provided new opportunities for brand endorsement strategy and gradually become an innovative trend. As the effectiveness of influencer endorsement is rooted in parasocial relationships, it remains unknown whether this effect can be extended to virtual influencers. With a focus on endorsement, the current study investigated the impact of human-like virtual influencers on parasocial relationships and their further outcomes. Two scenario-based experiments spanning two cultural contexts (Studies 1 and 2) provided sufficient evidence that users build weaker parasocial relationships with human-like virtual influencers due to the lower emotional engagement involved, which results in lower endorsement effectiveness in terms of brand interest and purchase intention. Moreover, a novel moderation role of influencer post modality (image vs. video) was found to attenuate the negative effect of human-like virtual influencers. The findings contribute to enhancing the understanding of users' relationship building with virtual influencers, and offer suggestions for influencer endorsements and communication strategies for virtual influencers.

1. Introduction

The practice of brand endorsement, wherein brands leverage the influence of well-known individuals for promotional purposes (Derdenger et al., 2018), has evolved from highlighting traditional celebrities to ordinary individuals as social media platforms have grown in use. By sharing aspects of their daily lives, opinions, and product experiences, ordinary individuals have begun to effectively influence and reach a wide audience on the Internet, taking on the role of so-called influencers (Bu et al., 2022). Unlike traditional celebrities or well-known public figures who already have achieved fame via traditional mass media, influencers gain their popularity through online social media interaction with their audiences. Due to the frequent two-way interactions, audiences are more likely to develop certain emotional feelings for influencers, including emotional engagement and parasocial relationships (PSRs), increasingly relying on influencers in their decision-making process. Influencers thus offer great marketing potential for brands and advertisers (Cheung et al., 2022; Farivar and Wang, 2022; Yuan and Lou, 2020). Indeed, brand managers have been found to be making $5.78 ROI for every $1 spent on influencer endorsement (Geyser, 2022).

In addition to human influencers, virtual influencers have been actively introduced as brand endorsers leveraging the advancement of artificial intelligence (AI) technology (Sokolova and Kefi, 2020). Prominent virtual influencers such as Lil Miquela, Shudu, and Imma on Instagram have endorsed brands such as KFC, Dior, Prada, and more (Molenaar, 2022), and the co-existence of virtual and human influencers in brand endorsement has continued to develop. Through AI technology, virtual influencers present themselves with extremely human-like appearances, audio-visual features, and engage in social activities that involve real-time interaction with multiple audiences simultaneously. Accordingly, virtual influencers are expected to yield similar impacts as human influencers with regard to arousing emotional connections and strengthening relationships in a more cost-effective manner.

To date, literature on virtual influencers' effectiveness has primarily centered on discussion related to their social interaction and has yet to reach consistent conclusions. Some research has reported a tendency among users to exhibit a higher frequency of interactive behaviors (i.e., likes or comments) with virtual influencers than with human influencers (Arsenyan and Mirowska, 2021). Conversely, it is imperative to acknowledge that interactions with virtual influencers can evoke distinct emotional responses (i.e., doubt and anxiety) and uncertainties

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among users due to feelings of uncanny (Arsenyan and Mirowska, 2021; Sands et al., 2022; Thomas and Fowler, 2021). Given virtual influencers' characteristics of fake authenticity, users may find themselves trapped in a state of uncertainty and uncanny regarding their true nature. However, existing studies remain on the identity skepticism discussion, neglecting the information conveyed by virtual influencers and the further impact in marketing outcomes. Virtual influencers, distinguished by their distinctive attributes in appearance and performance, are believed to engage in endorsements characterized by a unique pattern distinct from that of human influencers (Alboqami, 2023; B. Li et al., 2023). Hence, there is a scope for deeper investigation into the effectiveness of human-like virtual influencers in brand endorsement, both for academic and marketing purposes.

To our knowledge, a core function and purpose of influencers in endorsement is their ability to evoke emotional engagement and foster PSRs with their audiences (Stein et al., 2022). Given the limited current research on virtual influencers, our research extends the existing literature on influencers by investigating this new source – human-like virtual influencers – and, using parasocial relationship theory as a starting point, explores how audiences respond to human-like virtual influencers in comparison to human influencers on social media in terms of endorsement effectiveness. We argue that human-like virtual influencers, as advanced AI avatars which present similarly to human influencers, might elicit changes in audience behavior responses and carry-on outcomes. Through two experiments which spanned two cultural contexts, we unveiled the potential impact of human-like virtual influencers in the realm of brand endorsement. Our interpretation of these outcomes is grounded in the identification of underlying mechanisms, specifically, the roles of emotional engagement and PSRs as serial mediators. Additionally, we have identified a managerially pertinent moderator, namely post modality (image vs. video), which exerts a moderating influence on the brand endorsement effectiveness of human-like virtual influencers. These findings carry profound implications for advertisers, offering guidance on whether and how to strategically leverage virtual influencers on social media platforms. Furthermore, the findings hold relevance for brands that seek endorsement from virtual influencers, providing insights into how such influencers can impact social relationships in advertising strategies.

There were several rationales for conducting this research. First, we aimed to extend influencer literature by exploring a new source – virtual influencers. Previous influencer literature has examined a number of different influencer characteristics (i.e, perceived attractiveness, perceived expertise, and celebrity degree) that may influence outcomes (Janssen et al., 2022; Lou and Yuan, 2019; Masuda et al., 2022), however, there is limited research examining the source differences of influencers, particularly of virtual influencers, and their effectiveness in brand endorsement. Second, we aimed to contribute to the literature on brand endorsement and PSRs by investigating how influencer type impacts brand endorsement effectiveness. Depite virtual influencers garnering high levels of engagement (e.g., likes, comments) (Arsenyan and Mirowska, 2021), these interactions are related to their identities and do not appear to be effectively converted into brand endorsement effectiveness. This research extends the investigation on brand endorsement effectiveness through parasocial relationship theory and expands our understanding of the potential underlying psychological influences (i.e., emotional engagement, PSRs) to a non-human influencer context. Furthermore, we also attempted to explore the importance of post modality in order to contribute to communication strategy, which is a novel but neglected variable. By identifying the moderating effect of post modality, this research also contributes to the body of knowledge on parasocial relationship theory by complementing our understanding of the potential boundary of post-modality as an exposure activity.

2. Literature review and hypothesis development

2.1. Influencers as brand endorsers

Influencers are a new type of endorsers, individuals who share posts, blogs, and opinions on social media, and are able to shape their audience's attitudes (Ki et al., 2020). With a substantial following, influencers are able to assert their influence by providing their audience with information and relationship building (Farivar et al., 2021). Research has proven influencers' effectiveness in brand endorsement, benefiting both the endorsed brands and the influencers themselves. On the one hand, endorsement by influencers has been shown to generate favorable brand attitudes (Munnukka et al., 2019), as well as increased purchase intention (Stubb and Colliander, 2019) and brand recall (Boerman, 2020). On the other hand, through their genuine endorsements, influencers are shown favorable attitudes and build their followers (Janssen et al., 2022) while monetizing their fame (Liljander et al., 2015), which in turn can further help them in their brand endorsement. In addition, increasing research has found that influencers may in fact be even more effective in brand endorsement than traditional celebrities, especially for the younger generation (Djafarova and Rushworth, 2017). For example, influencers have been shown to generate a more significant impact on purchase decisions than traditional celebrities due to their higher perceived similarity, credibility, and relatability (Schouten et al., 2020).

Recent advances in AI have triggered a shift in the realm of influencers, with an innovative new type of influencer known as virtual influencer (also called AI influencer), partaking in brand advocacy (Sands et al., 2022; Thomas and Fowler, 2021). Virtual influencers are digitally designed characters that typically possess a realistic appearance and figure in a computer-generated way (Miao et al., 2022). Though they can vary strongly in terms of their visual fitness, virtual influencers can be human-like, cartoon-like, or anime-like, and engage in activities similar to human influencers on social media. Virtual influencers also imitate human activities, such as sharing their daily life activities and being involved in endorsing brands, however, they are able to respond to multiple audiences with tailored content simultaneously leveraging AI technology. Existing research on virtual influencers is limited in social interaction and has not achieved consistent conclusions due to the true nature of virtual influencers. Positive wise, virtual influencers can generate positive engagement (such as following intention) and create a sense of personalization just like human endorsers (Sands et al., 2022; Thomas and Fowler, 2021); negative wise, virtual influencers have also received significantly fewer positive reactions doubting their identities and source trust from consumers when compared to human influencers (Arsenyan and Mirowska, 2021; Sands et al., 2022) (Please see Appendix D for selected literature review table). While numerous virtual influencers boast realistic human-like appearances and attractiveness that have the potential to amplify their effectiveness (Kim and Park, 2023), however, it is their identities that represent the primary concern for users. The ontological nature of this new type of virtual entity is the source of current debate (Stein et al., 2022), however, little discussion has been made on their related activities and their performance, such as brand endorsement. In light of these gaps, this research focuses specifically on the comparison of human-like virtual influencers and human influencers in the role of brand endorsers.

2.2. Parasocial relationship theory

Parasocial relationship theory was originally proposed by Horton and Richard Wohl (1956) to explain the formation of social relationships between media characters and their audiences. PSRs refer to the relationships between users and personas (i.e., mediated representations of presenters, celebrities, or characters) with an illusion of intimacy for these "reciprocal" interpersonal relationships which are in fact inherently a one-way relationship (Giles, 2002; Hartmann and Goldhoorn,

2011; Sokolova and Kefi, 2020). Such relationships can evolve to the point where users begin to view the mediated others as "real friends" (Stern et al., 2007), however, it is actually an altruistic, self-established human instinct to form attachments with others. Meanwhile, the personas involved in these PSRs being constructed might be unaware of these relationships and unknowingly influence them.

In the context of social media influencers, influencers are media characters that bridge the gap between brands and users, actively fostering the formation of PSRs with their followers (Bi and Zhang, 2022). As influencer posts are updated in real-time, audience users are able to further their illusions that they "know" the influencers (Sokolova and Kefi, 2020). Moreover, the ability to "like", comment, and retweet posts instantly also foster a sense of direct communication with the influencers, especially when these receive responses from the influencer. In response to experiencing this kind of virtual bidirectional interaction, users will build strong PSRs with the influencers (Hu et al., 2020).

Within this one-way relationship, a wide range of source-related factors (i.e., source characteristics and source activities) of influencers have been identified that influence PSRs. On the one hand, source characteristics of influencers can affect users' impressions and subsequent interaction experiences similar to interpersonal interactions. For example, interpersonal attraction, encompassing physical and social attractiveness, can lead to the formation of first impressions and initiate social relationships, which are positively associated with PSRs (Tukachinsky and Stever, 2019). The perceived homophily of influencers can make interactions smoother and more enjoyable, which also contributes to PSRs (Tukachinsky et al., 2020). Source activities also play a significant role in promoting PSRs. For instance, perceived interactivity transfers a feeling of direct two-way communication and interaction, which fosters PSRs. Furthermore, exposure activities, such as self-disclosure, openness, and expression of affection, can facilitate PSRs through the enhancement of a mutual understanding of one another (Labrecque, 2014). Through PSRs, influencers generate source trustworthiness and perceived expertise, which together empower them to assert specific persuasive influences on their audiences in numerous ways including enhancement evaluation of the endorsement, brand attitudes, and brand affection (Breves et al., 2021; Lim et al., 2020).

The effectiveness of PSRs is largely dependent on the source itself. While present research has focused on source-related factors, the majority of the focus remains on human influencers, with the exception of a limited number of studies that have explored virtual influencers, a fictional source presenting a realistic human-like appearance and activities. While research does exist investigating parasocial interactions (e.g., likes, comments, attitudes) between real influencers and virtual influencers (Arsenyan and Mirowska, 2021), the results remain inconsistent showing that virtual influencers arouse no difference or even higher interaction with real influencers. However, parasocial interactions refer to episodic media exposure experiences, while PSRs refer to long-term bonds that users develop with influencers, which may form during exposure but can extend beyond the end of the exposure, and can be developed with or without parasocial interactions (Dibble et al., 2016; Rubin et al., 1985; Tukachinsky and Stever, 2019). Limited qualitative study has reported that users may have weaker parasocial relationships with virtual influencers (Lou et al., 2022). Building upon this understanding, the current research thus moves forward in examining the ongoing advertising outcomes of PSRs on endorsement effectiveness, taking a unique perspective on the source of the influence by examining brand endorsement outcomes via PSRs.

2.3. Influencer type and PSRs

According to parasocial relationship theory, source characteristics are found to impact PSRs significantly, such as source credibility, source humanness, and source fairness (Whang and Im, 2021; Yuan and Lou, 2020). Introduced by Giles (2002), PSRs are believed to be automatically elicited through cues related to human characteristics. For example, artificial entities, including cartoon and anime characters, virtual avatars, and bodiless chatbots, have been proven to generate lower parasocial responses in comparison to human entities (e.g., Ramasubramanian and Kornfield, 2012; Youn and Jin, 2021). Voice assistants are shown to be perceived to be more humanlike than information presented on a website, and are more likely to encourage stronger PSRs with consumers (Whang and Im, 2021). Giles (2002) also argued that, compared to actual humans, fictional characters might require a greater leap of faith from users when it comes to PSRs building.

Following the above aspects of parasocial relationship theory, influencer type (i.e., virtual or human) should play a role in the development of PSRs. Although human-like virtual influencers are programmed to imitate human characteristics with both appearance and intelligence realism (Miao et al., 2022), they are not real human beings, but rather virtual characters with an artificial appearance (Stein et al., 2022). As artificial entities are perceived to lack genuine human mental capacities in terms of sensing (e.g., touch, smell; Gray and Wegner, 2012), users have been shown to perceive them as being less knowledgeable and empathetic, and are less likely to interact with AI (Hill et al., 2015; Luo et al., 2019; Mou and Xu, 2017). Research has shown that human-like virtual influencers receive significantly fewer positive reactions and source trust from users when compared to human influencers (Arsenyan and Mirowska, 2021; Sands et al., 2022). It is thus reasonable to assume that users would have a lower inclination to build PSRs with human-like virtual influencers than with human influencers, due to their artificial nature. Therefore, the following hypothesis was proposed.

H1. PSRs with human-like virtual influencers will be weaker than with human influencers.

2.4. The mediation role of emotional engagement

If it is the case that the PSRs with human-like virtual influencers are weaker than with human influencers, we then proposed that these weakened PSRs stem from the decrease in emotional engagement with human-like virtual influencers.

A wide range of existing literature has suggested that emotional engagement toward artificial entities and real humans differs between users (Perez-Vega et al., 2021). As emotional engagement involves emotional connection and expression in interpersonal interaction (Hilvert-Bruce et al., 2018; Lim et al., 2015), we will elaborate on these two emotional responses. In terms of emotional connection, research has generally found that users have a lower tendency to build emotional connections with artificial entities as they assume that these characters lack humanness and empathy. For example, chatbots are viewed as depersonalized and devoid of emotion, resulting in users' increased perception of their lack of uniqueness (Longoni and Cian, 2022) and empathy (Luo et al., 2019).

In terms of emotional expression, users have also been found to display less emotion when engaging with artificial entities in online communication. For instance, users exhibit greater levels of extroversion, conscientiousness, and self-disclosure, and employ a richer vocabulary when communicating with human agents as compared to chatbots (Hill et al., 2015; Mou and Xu, 2017). Similarly, human-like virtual influencers have been shown to evoke more negative emotions such as skepticism and anxiety toward their identity from users in social interactions regardless of the post information (Arsenyan and Mirowska, 2021; Sands et al., 2022). Accordingly, in the field of brand endorsement, users' concern remains on virtual influencers' nature identity rather than brand endorsement information. Therefore, users are assumed to assert less emotional connections and emotional expressions toward virtual influencers, lacking the emotional arousal process in response to their endorsement. Human-like virtual influencers, as artificial entities, were therefore hypothesized to arouse lower levels of emotional engagement than human influencers.

H2a. Emotional engagement toward human-like virtual influencers will be weaker than toward human influencers.

Following parasocial relationship theory, emotional interaction plays a critical role in determining the development of PSRs (Deng et al., 2022). Emotional engagement refers to one's intrinsically motivated involvement and enthusiasm for other individuals, which can satisfy the need for PSRs (Kowert and Daniel, 2021). Emotional responses toward media characters have been recognized as a one-sided reaction, which can also foster one-way attachment, such as PSRs (Deng et al., 2022).

Social media influencers interact with users through comments (Reinikainen et al., 2020). Emotional connection and emotional expression can both be easily reflected in comments. The feeling of emotional connection leads to users' intrinsically motivated involvement and enthusiasm for the influencers, leading users to be more actively engaged and bonded with the influencers. Furthermore, emotional expression in interactions can facilitate intimate feelings and lead to subsequent relationship building. Evidence from previous research has revealed that users who experience a high level of emotional engagement are more likely to engage in PSRs in the context of live streaming and social TV (Lim et al., 2020; Shin, 2016). Therefore, we hypothesized that, when users feel more emotional engagement with an influencer, they will show a higher tendency to build PSRs with that influencer.

H2b. Emotional engagement toward influencers is positively associated with PSRs.

2.5. Influencer type and brand endorsement effectiveness

Parasocial relationship theory highlights the persuasiveness effect of PSRs. For example, PSRs between influencers and users have a positive impact on brand endorsement in terms of attitudinal (i.e., brand attitude, brand attachment) and behavioral (i.e., purchase intention, word-of-mouth) responses among users (Hwang and Zhang, 2018; Ki et al., 2020). Specifically, when users form strong PSRs with influencers, they perceive brand endorsements as recommendations from friends and are more likely to exhibit a positive attitude toward the brands and spend more time focusing on posts by the influencer (Casaló et al., 2020), thereby enhancing their willingness to purchase and recommend the endorsed brands (Labrecque, 2014; Vazquez et al., 2020).

Given that users build weaker PSRs with human-like virtual influencers than they do with human influencers, the endorsement effectiveness of human-like virtual influencers should reduce correspondingly (Chung and Cho, 2017). Therefore, we proposed that reduced PSRs induced by human-like virtual influencers will negatively impact brand endorsement effectiveness in terms of brand interest and purchase intention (Lefebvre and Cowart, 2021), hypothesizing the following.

H3. Human-like virtual influencers are negatively associated with brand endorsement effectiveness (i.e., brand interest and purchase intention) compared with human influencers.

2.6. The moderating role of influencer post modality

At the core of influencer endorsement on social media is information sharing, which is the primary activity of influencers. According to parasocial relationship theory, the exposure activity of influencers, such as self-information exposure, is an important antecedent of PSRs. This information sharing works as a form of exposure, and tends to also impact users' emotional responses and PSRs toward influencers (Labrecque, 2014). Empowered by technology, information sharing in an influencer's post can have multiple modalities, namely text, images, and videos (Feng et al., 2021). Images and videos are the main modalities of influencer posts, both involving a text description (Feng et al., 2021), thus we focused on the comparison of images and videos in our research.

For users, the information richness of influencers' posts varies across post modalities, consisting of both endorsement information and influencer status updates, which can bring about a heightened exposure perception in terms of self-information and emotion expression. On the one hand, self-related information, such as the influencer's feelings or thoughts, tends to be perceived more strongly when shared in videos than in images (Huang et al., 2008; J. Li et al., 2023; Ramirez et al., 2004). The richness of self-information can enrich the "human element" and create a more detailed portrait of the influencer, resulting in posts which are more likely to arouse users' emotional responses (Stein et al., 2022). Accordingly, the heightened perceived self-information disclosed by the influencer places users in an illusionary situation that they are building emotional connections with the influencer and should respond via emotional expression in this two-way interaction. On the other hand, as video has been proven to provide superior richness over images in terms of their metrics of multiplicity of cues and added heuristic attributes, information transmitted by video can be more salient, vivid, and provide deeper sensory richness (Lee et al., 2021), which will then arouse users' mutual understanding and attract users' emotional engagement, subsequently further strengthening PSRs (Guadagno et al., 2013). Furthermore, the higher exposure perception brought about by videos leads to a perception or awareness of others in online environments (Kim, 2022), which creates an environment that is supportive for emotional expression, open interaction, and understanding construction, promoting emotional engagement and relationship building in users (Molinillo et al., 2018).

In the current research, despite the importance of influencer posts as informational content, this has not yet been examined as a moderator in previous influencer literature. As video results in an increased impact of influencers' exposure activities such as self-information and emotional sharing as compared to images (Molinillo et al., 2018; Yoo and Alavi, 2001), it further promotes emotional engagement and PSRs from users. For human-like virtual influencers, video can better present their human-like features and bodily motions, more so than static images, making them seem more like real humans. The information richness conveyed via video enhances the goodness perception of human-like virtual influencers, which increases users' tendencies to become more emotionally engaged and build PSRs with them. Thus, we proposed the following hypothesis (see Fig. 1).

H4. Brand endorsement effectiveness is lower in human-like virtual influencers compared to human influencers when post modality is image. This effect will be attenuated when post modality is video.

3. Study 1

The purpose of Study 1 was to seek support for H1 under a controlled experimental setting and explore the details of the mechanism effect. Study 1 manipulated the influencers' type and sought evidence for the impact of whether influencer type (i.e., human-like virtual influencer vs. human influencer) impacted brand endorsement effectiveness through emotional engagement and PSRs by measuring perceptual outcomes (i.e., brand interest and purchase intention).

3.1. Study design

A between-subjects experiment (influencer type: human-like virtual influencer vs. human influencer) was conducted. 260 samples (age range = 18–60, M_{age} = 31.03, 44.62% male) were recruited in May 2022 via Credamo,[1] a professional Chinese online survey platform, in exchange for monetary compensation. Participants were assigned randomly to one of the influencer conditions. Then, participants were asked to imagine that they were browsing their social media page and an

[1] www.credamo.com.

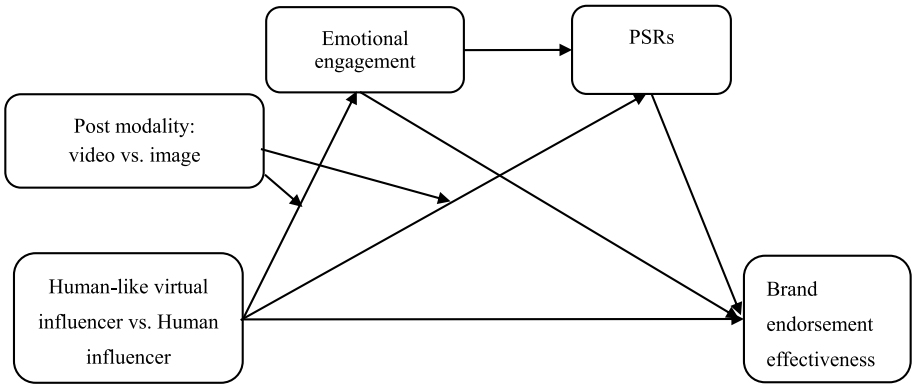


Fig. 1. Theoretical framework.

influencer appeared in their feed. Next, an influencer's homepage featuring name, image, a brief introduction, and recent updates was shown to participants. Following Thomas and Fowler (2021) methodology, we manipulated the influencer conditions by manipulating their names. Under the human influencer condition, the influencer called "Lena" with a brief introduction displays "life blogger", while under the human-like virtual influencer condition, the influencer called "Virtual_Lena" with a brief introduction displays "virtual influencer_life blogger". Both conditions used the same human-like image and shared updates to avoid possible interfering factors such as attractiveness, similarity, and celebrity worship. After reading the background material, participants indicated their perception of the influencer by completing a manipulation check. After that, participants were shown a post from the influencer, accompanied by a statement that she is cooperating with Caltin, a healthy yogurt brand. Brand interest, purchase intention, emotional engagement, and PSRs were then measured as outlined in the following section. The original English measurement scale was translated into Chinese and then back into English by professional translators to ensure measurement equivalence.

3.2. Measurements

Manipulation check for the influencers perception was performed using a seven-point semantic-differential scale as developed by Mozafari et al. (2021) which asked, "Please rate whether you think you viewed a virtual influencer or a human influencer" anchored by virtual influencer (1) and human influencer (7).

Emotional engagement was measured using a scale developed by Lim et al. (2015), with the scale including items such as "I expressed my feelings about the Lena/Virtual_Lena under her posts." Cronbach's alpha was 0.82, indicating satisfactory reliability for emotional engagement.

PSRs was measured using a scale from Lim et al. (2020), including "If Lena/Virtual_Lena would appear live in another post, I would view that post". Cronbach's alpha was 0.81, indicating satisfactory reliability for PSRs.

Brand endorsement effectiveness was measured by assessing participants' brand interest and purchase intention. Brand interest was measured using four items from the scale developed by Machleit et al. (1993), with a sample question being, "I am intrigued by Caltin". Cronbach's alpha was 0.84, indicating satisfactory reliability for product attitude. Purchase intention was measured using a scale from Janssen et al. (2022) with one item, "The next time you are looking to purchase this type of product, how likely are you to buy Caltin's products?".

3.3. Results

Manipulation check. The manipulation check for influencers was significant at $p < .001$, with participants in the human-like virtual influencer group perceiving they are viewing virtual influencer more than those in the human influencer group ($M_{VI} = 3.33$, $SD = 1.07$; $M_{HI} = 5.10$, $SD = 0.92$; $t = -14.42$).

Effect of influencer type. A multivariate analysis was conducted to test the differences between virtual influencers and human influencers in terms of brand endorsement effectiveness, emotional engagement, and PSRs. The results indicated a significant effect of influencer type (F (4, 255) = 13.08, p < .001, $\eta_p^2 = 0.17$) on emotional engagement, PSRs, and brand endorsement effectiveness. Specifically, a t-test was utilized to report impact of influencer type on the three dependent variables. Participants hold a lower level of emotional engagement toward virtual influencer than human influencer ($M_{VI} = 4.55$, $SD = 1.16$; $M_{HI} = 5.34$, $SD = 1.06$; $t = -5.71$, $p < .001$), demonstrating H2a. Besides, the participants also showed a lower level of PSRs toward virtual influencer than human influencer ($M_{VI} = 5.32$, $SD = 1.08$; $M_{HI} = 5.76$, $SD = 0.84$; $t = -3.74$, $p < .001$), supporting H1. For the endorsement effectiveness, compared to human influencer, virtual influencer resulted in lower endorsement effectiveness in terms of brand interest ($M_{VI} = 5.16$, $SD = 1.17$; $M_{HI} = 5.44$, $SD = 0.99$; $t = -2.08$, $p < .05$) and purchase intention ($M_{VI} = 4.82$, $SD = 1.34$; $M_{HI} = 5.54$, $SD = 1.22$; $t = -4.51$, $p < .001$), supporting H3.

To demonstrate the robustness, we further conducted an ANCOVA analysis with participants' gender, age, level of education, and income as co-variates. The analysis revealed a similar interactive effect of influencer type on emotional engagement (F (1, 259) = 31.26, $p < .001$, $\eta_p^2 = 0.11$), PSRs (F (1, 259) = 13.60, $p < .001$, $\eta_p^2 = 0.05$), brand interest (F (1, 259) = 3.16, $p < .05$, $\eta_p^2 = 0.01$), and purchase intention (F (1, 259) = 20.43, $p < .001$, $\eta_p^2 = 0.07$).

Mediation effect of emotional engagement. The mediation analysis (Hayes, 2018, Model 4) for SPS with 5000 bootstrap samples demonstrated that emotional engagement (indirect effect = −0.40, SE = 0.09, 95% CI = [−0.59, −0.24]) mediated the relationship between influencers and PSRs. Thus, H2b was supported.

Serial mediation. We then employed a set of serial mediation models (Hayes, 2018; Model 6 for serial mediation) for SPSS using 5000 bootstrapped samples. We used emotional engagement and PSRs as sequential mediators, and brand endorsement as the outcome. This model revealed that users in the human-like virtual influencers condition exhibited decreased emotional engagement and PSRs, resulting in lower brand interest (indirect effect = −0.26, SE = 0.07, 95% CI = [−0.40, −0.14]) and purchase intention (indirect effect = −0.18, SE = 0.07, 95% CI = [−0.33, −0.07]; see Figs. 2 and 3 for path models with estimated coefficients).

3.4. Discussion

Study 1 reveals important managerial implications with regard to the use of human-like virtual influencers, specifically suggesting potential drawbacks to using human-like virtual influencers, by revealing the actual user perceptions of human-like virtual influencers. Given that we

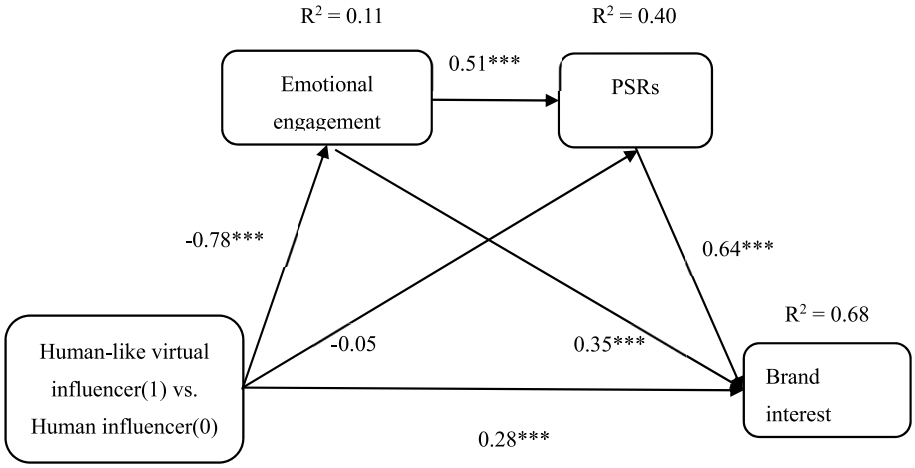


Fig. 2. The mediating effect test for brand interest.

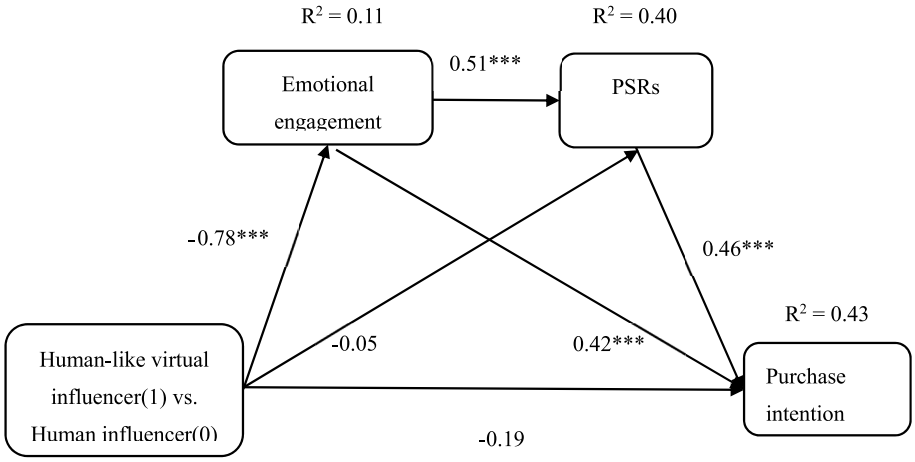


Fig. 3. The mediating effect test for purchase intention.

focused on the source difference between human-like virtual influencers and human influencers on brand endorsement effectiveness, the remaining studies focused further on differences in post content that may moderate the effect. Furthermore, we expanded our influencer types to include fashion influencers and male characteristics, as well as to include samples from Western countries to enhance the generalizability of our findings.

4. Study 2

There were two main goals of Study 2. First, while Study 1 identified the mediation effect of human-like virtual influencers, the current study examined whether the findings observed thus far were generalizable to different national markets as well as across specific influencer characteristics (i.e., gender, portrait, category). Second, we wanted to extend the current findings of the mediation effect of emotional engagement by introducing the moderation effect of post content. Accordingly, we proposed that post content (i.e., video vs. image) can impact users' perception toward influencers and therefore impact brand endorsement. To do so, we predicted that when influencers share video content, users' emotional engagement is enhanced, thus leading to users having a better attitude toward the product being endorsed by the influencer.

4.1. Study design

554 samples (age range = 18–60, M_{age} = 31.30, 58.12% male) participated in this study via Prolific in May 2022 in exchange for monetary compensation. The study consisted of a 2 (Influencers: human influencer vs. human-like virtual influencer) x 2 (Post modality: video vs. image) between-subjects design. First, participants were asked to imagine that they were browsing their social media page and came across an influencer in their feed. Next, participants were shown the influencer's home page which included their name, image, a brief introduction, and recent updates. In the human influencer condition, the influencer was named "Simon", and his brief introduction showed that he was a "fashion blogger" displaying a human portrait. In the human-like virtual influencer condition with source disclosure, the influencer was named "Virtual_Simon", and their brief introduction showed that they were a "virtual influencer_fashion blogger", displaying a similar portrait to the image used in the human influencer condition but with fewer facial wrinkles. The manipulation of the influencers had been successfully validated by a pretest in terms of both influencer perception and appearance similarities. Further, users were presented with one specific post of the influencer endorsing a niche clothing brand named "Platii". In the image condition, the endorsement post consisted of some images from the different perspectives of influencers wearing the clothes, while in the video condition, the endorsement post was a 15-s

video presenting the influencer dressed in a full range of their clothing, presenting a dynamic view of the outfits. Each scenario was also presented with a similar image of the human-like influencer and the same recent updates. After reading the background material, participants were asked to complete the measurement items for the manipulation check, perceived emotional engagement, PSRs, and brand endorsement effectiveness, which included brand interest and purchase intention. The measurement scales were the same as was used in Study 1 (please see the detailed construct reliability and validity in Appendix C).

4.2. Results

Manipulation check. The manipulation check for influencers was significant at $p < .001$, with participants in the human-like virtual influencer group perceiving they are viewing virtual influencer more than those in the human influencer group ($M_{VI} = 3.58$, $SD = 1.12$; $M_{HI} = 5.27$, $SD = 0.88$; $t = -19.87$, $p < .001$).

Effect of influencer type. A multivariate analysis was conducted to test the differences between virtual influencers and human influencers in terms of brand endorsement effectiveness, emotional engagement, and PSRs. The results indicated a significant effect of influencer type (F (4, 549) = 6.83, p < .001, $\eta_p^2 = 0.05$) on emotional engagement, PSRs, and brand endorsement effectiveness. Similar to the result of Study 1, the t-test showed that compared to human influencer group, the participants in virtual influencer group showed a lower level of emotional engagement ($M_{VI} = 5.08$, $SD = 1.23$; $M_{HI} = 5.43$, $SD = 1.06$; $t = -3.60$, $p < .001$), PSRs ($M_{VI} = 5.50$, $SD = 0.73$; $M_{HI} = 5.83$, $SD = 0.73$; $t = -5.19$, $p < .001$), brand interest ($M_{VI} = 5.24$, $SD = 0.93$; $M_{HI} = 5.54$, $SD = 0.88$; $t = -3.89$, $p < .001$), and purchase intention ($M_{VI} = 5.53$, $SD = 1.25$; $M_{HI} = 5.77$, $SD = 1.08$; $t = -2.42$, $p < .05$).

To demonstrate the robustness, we further conducted an ANCOVA analysis with participants' gender, age, level of education, and income as co-variates. The analysis revealed a similar interactive effect of influencer type on emotional engagement (F (1, 553) = 12.82, p < .001, $\eta_p^2 = 0.02$), PSRs (F (1, 553) = 29.15, p < .001, $\eta_p^2 = 0.05$), brand interest (F (1, 553) = 16.84, p < .001, $\eta_p^2 = 0.03$), and purchase intention (F (1, 553) = 5.37, p < .05, $\eta_p^2 = 0.01$).

Serial mediation. We employed a mediation model (Hayes, 2018; Model 6 for serial mediation) for SPSS using 5000 bootstrapped samples, indicating that human-like virtual influencers lead to lower levels of both brand interest (indirect effect = −0.06, SE = 0.02, 95% CI = [−0.10, −0.03]) and purchase intention (indirect effect = −0.06, SE = 0.02, 95% CI = [−0.10, −0.03]) through serial mediation of emotional engagement and PSRs.

Moderation analysis. Finally, we conducted a bootstrap analysis to estimate the multistep moderation model (Hayes, 2018; Model 84 for moderation) for SPSS with 5000 bootstrapped samples. The results showed that the indirect effect of influencer type on brand interest (video indirect effect = −0.09, SE = 0.05, 95%CI = [−0.19, 0.01]; image indirect effect = −0.16, SE = 0.06, 95%CI = [−0.28, −0.06]) and purchase intention interest (video indirect effect = −0.08, SE = 0.05, 95%CI = [−0.17, 0.01]; image indirect effect = −0.15, SE = 0.06, 95% CI = [−0.28, −0.05]) via emotional engagement was significant only in the image condition. Besides, the indirect effect of influencer type on brand interest (video indirect effect = −0.05, SE = 0.04, 95%CI = [−0.13, 0.02]; image indirect effect = −0.12, SE = 0.04, 95%CI = [−0.20, −0.06]) and purchase intention interest (video indirect effect = −0.05, SE = 0.04, 95%CI = [−0.13, 0.03]; image indirect effect = −0.12, SE = 0.04, 95%CI = [−0.21, −0.05]) via PSRs was also significant only in the image condition.

While replicating our effects from the previous studies, the moderating model revealed that post modality attenuated the negative effect of human-like virtual influencers on brand endorsement. The negative endorsement effect of human-like virtual influencers was significant only in the image condition: brand interest (video indirect effect = −0.04, SE = 0.03, 95%CI = [−0.10, 0.01]; image indirect effect = −0.08, SE = 0.03, 95%CI = [−0.14, −0.03]); purchase intention (video indirect effect = −0.04, SE = 0.02, 95%CI = [−0.09, 0.01]; image indirect effect = −0.08, SE = 0.03, 95%CI = [−0.13, −0.03]). Accordingly, H4 was supported. The path model with estimated coefficients is displayed in Figs. 4 and 5.

4.3. Discussion

Study 2 identified a relevant boundary condition for the effect of influencers on brand endorsement from a content-related perspective. The results showed that influencers' brand endorsement effectiveness could be enhanced when the post content was a video rather than an image through enhancement of emotional engagement and PSRs. This finding further clarified our theory extension from parasocial relationship theory, highlighting the importance of information richness in influencer posts. Our findings also provide further empirical evidence for the importance of emotional engagement as an intermediate and mediating outcome of brand endorsement effectiveness. Overall, the results of Study 2 contribute to the broader understanding of influencer marketing by shedding light on the importance of content-related factors and the mediating role of emotional engagement and PSRs in the effectiveness of brand endorsement.

5. Conclusion and implication

5.1. General discussion

Firstly, as influencer advertising continues to gain prominence as a marketing strategy, the findings from our current studies offer valuable guidance for brands and advertisers. Traditionally, research has predominantly focused on understanding the effects of source characteristics of human influencers, however, evolving technology has given rise to virtual influencers, adding new dimensions to the source itself (Miao et al., 2022). Concurrently, while existing research on virtual influencers has primarily explored their performance in the domain of social interaction, limited attention has been paid to their future marketing outcomes. This research addresses this gap through two experiments spanning two cultural contexts, showing that human-like virtual influencers negatively impact brand endorsement through weakened emotional engagement and PSRs, revealing no cultural differences in virtual influencers' effectiveness. Additionally, influencers' post modality (i.e., video vs. image) was found to attenuate the negative brand endorsement effectiveness of human-like virtual influencers.

Secondly, we applied our findings to assess their implications for marketing outcomes and endorsement effectiveness within the framework of parasocial relationship theory. Building upon current arguments regarding the effect of virtual influencers, our findings address the gap of research in understanding the tangible marketing outcomes resulting from these interactions. Although previous research has found that virtual influencers result in more interactions such as comments and likes and more negative emotions (Arsenyan and Mirowska, 2021), however, these interactions remain on their identity issue rather than their posted information. Our results showed that users show a lower level of emotional engagement toward virtual influencers in terms of their brand endorsement, which leads to lower PSRs and negative endorsement outcomes in terms of brand interest and purchase intention. One interesting point is that there exists a cultural difference in the path effect. Compared to China, influencer type can result in a significant difference in PSRs in Western countries. This can be interpreted by the collective culture of the Chinese, which emphasizes collective identity over individual identity compared to Western countries, and as a result, Chinese are less inclined to form deep, one-sided connections (Du et al., 2023).

Thirdly, regarding the moderators, we brought out the managerial moderators on post modality, which we tested in Study 2. We found that when human-like virtual influencers share video posts, emotional

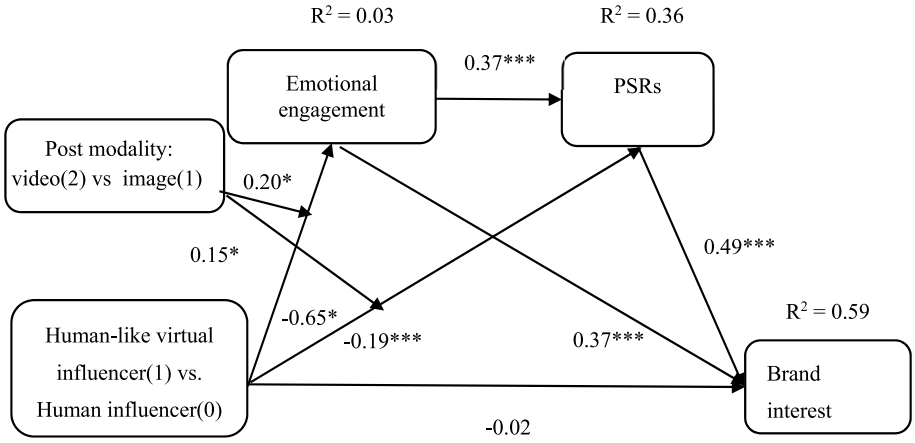


Fig. 4. The moderating effect test for brand interest.

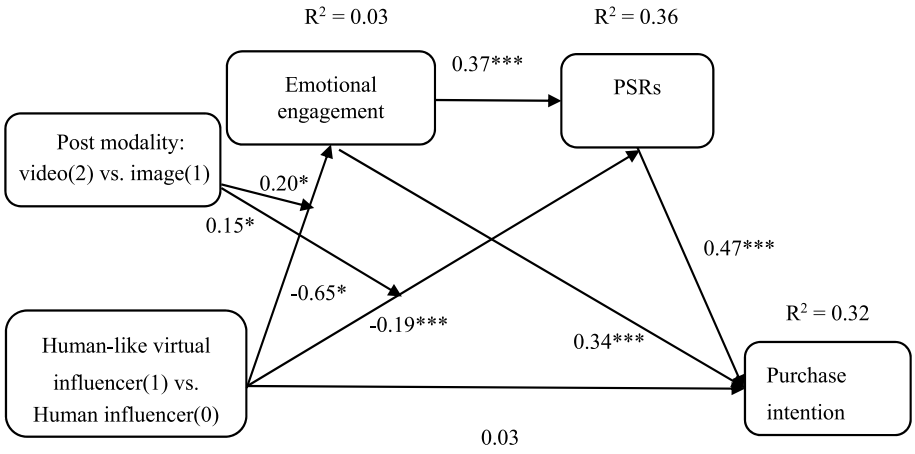


Fig. 5. The moderating effect test for purchase intention.

engagement and PSRs can be higher than with image posts, resulting in stronger PSRs and more favorable ad evaluations. When human influencers share video posts, the inclusion of the post modality did not impact either emotional engagement or the PSRs. These results suggest that influencers' post modality should be an essential consideration for advertisers and brands when planning exposure activity.

5.2. Theoretical contributions

This research enriches the existing research on influencer advertising by examining virtual influencers' endorsement effects from an emerging source perspective. It is important to note that previous research on influencers has primarily examined the factors and mechanisms associated with influencers' effectiveness, with the majority of focus being on human influencers' characteristics (e.g., attractiveness, homophily; Janssen et al., 2022; Masuda et al., 2022). Little research, to our knowledge, has explored the source of virtual influencers. Thus, we have extended the work of influencer advertising by introducing this new type of endorser, and investigating the possible endorsement effects of human-like virtual influencers.

Second, the current research contributes to the literature on brand endorsement effectiveness by identifying an underlying mechanism as suggested by parasocial relationship theory. While previous studies have highlighted the effectiveness of virtual influencers in improving interaction (i.e., likes or comments), these interactions are not always positive and concerning their identities (Arsenyan and Mirowska, 2021; Sands et al., 2022). Debate remains as to the effects of identity perception of virtual influencers, and there is a lack of research into following the marketing outcomes of the brand endorsement effectiveness of virtual influencers. This research attempts to address this gap by using parasocial relationship theory to investigate endorsement effectiveness in terms of brand interest and purchase intention. We find that emotional engagement and PSRs serially mediated brand endorsement effectiveness of influencer type. These findings not only extend the previous understanding of the impact of emotional engagement on PSRs to virtual characters (Lim et al., 2020), but also reinforce the importance of PSRs in contributing to marketing effectiveness.

Finally, this research introduces an empirically driven influencer communication moderation strategy and enriches influencer advertising from a content perspective. Prior research has explored strategies for improving user-influencer communication based on content characteristics, focusing primarily on message value, ad disclosure, and product-influencer congruence (Kim and Kim, 2020). This study introduces a critical but neglected content strategy to moderate the impact of human-like virtual influencers, and concludes that post modality with higher information richness, such as video, can attenuate users' low emotional engagement with human-like virtual influencers. Furthermore, these results add to the understanding of PSR theory by revealing the novel boundary of post modality.

5.3. Managerial implications

From a managerial perspective, this research offers valuable insights for brands and advertisers on whether to utilize influencers in their

marketing strategies. With the increasing presence of virtual influencers in brand endorsement, their effectiveness as endorsers demands closer scrutiny. Our findings provide initial evidence showing that human-like virtual influencers are not necessarily the best candidate for brand endorsement. According to our results, compared to human influencers, human-like virtual influencers lead to significantly lower emotional engagement as well as weaker PSRs, which together ultimately reduce the brand endorsement effect. Hence, brands and advertisers should prefer human influencers over virtual influencers, and be conscious of the limitations of virtual influencers as endorsers.

Furthermore, this study offers strategies on how to embrace virtual influencers in brand endorsements. Given the major concern of virtual influencers is their natural identity, a fluid and dynamic presentation can contribute to a more favorable attitude among users. According to our results, posts from human-like virtual influencers tend to be viewed more favorably when presented in video format rather than as static images, thereby enhancing users' emotional engagement. Consequently, a dynamic strategy, such as utilizing video posts and video-based platforms, should be considered when employing virtual influencers in marketing campaigns.

Moreover, this study also provides further guidance on future virtual influencers' design and strategies. Our findings underscore that the effectiveness of influencer endorsement hinges on users emotional engagement and PSRs. Notably, human-like virtual influencers tend to elicit lower emotional engagement and PSRs compared to human influencers. This highlights the importance of focusing on emotional capabilities of virtual influencers and their capacity to establish meaningful connections with their audience in the design and strategy development for virtual influencers moving forward.

5.4. Limitations and future research

Overall, the current research is an initial exploration into understanding the endorsement effect of virtual influencers. Our findings provide a solid starting point for future research, although several limitations do exist in the current studies. First, this study focused on human-like virtual influencers, but did not consider specific detailed features of these virtual influencers such as attractiveness and celebrity degree, which are source-related factors that have been shown to impact user attitudes with regard to human influencers (Janssen et al., 2022; Masuda et al., 2022). Future studies should consider the different effects of these characteristics when investigating virtual influencers and explore virtual influencers with other types of appearances as well, such as anime-like virtual influencers.

Second, this study aimed to examine the mechanisms of PSRs between users and different influencer types from a source-related perspective, however, the type of PSRs they established (i.e., friendship, exchange relationship) were neglected. And the emotional interaction such as valence and cognitive perception such as mental human-likeness and uniqueness requires further investigation. Going forward, we should expand virtual influencer research into the fields of customer service and entertainment, such as looking at virtual anchors or virtual employees. Future research should examine different user-virtual influencer relationships, valence interaction, and their carry-on impacts on other domains.

Furthermore, this study explored the moderation effect of post modality, providing preliminary proof of the effectiveness of the form of the endorsement message. However, we focused only on the comparison between videos and images, regardless of the detailed features of these modalities. Future studies should further investigate the features of these modalities, considering details such as image color or video length. As live streaming becomes the new standard of posting format, the effectiveness of virtual influencers in the context of live streaming should be explored promptly. Given the current study only included gender, age, education, and income as co-variates, future studies should involve social media platforms, as well as social media usage (e.g., time and frequency) in the measurement. Besides, as social media platforms affordances vary, future studies are encouraged to examine user reactions to virtual influencers across a variety of social media platforms based on affordances (e.g., Instagram, YouTube).

Finally, virtual influencers are a current trend in influencer marketing, and promise great potential for development and application. Virtual influencers will evolve over time and users' attitudes and perceptions will vary accordingly. Future research can conduct a time-series study on virtual influencers and trace subtle changes across time. Besides, given the limited activities of current human-like virtual influencers and technology restrictions, the current manipulation of virtual influencers origins from existing virtual influencers with fictitious names. Future research would benefit from the manipulation design or cooperation with existing virtual influencers to track real-time data.

Declaration of competing interest

The authors declare that they have no known competing financial interests that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A

Study Measures

Emotional engagement (Lim et al., 2015, 1 = Strongly Disagree, 7 = Strongly Agree)
I quoted from Lena/Virtual_Lena when she said something good or witty.
I expressed my feelings about the Lena/Virtual_Lena under her posts.
I posted my feelings in real-time online conversation or post comment with Lena/Virtual_Lena.
I posted my feelings when I liked/disliked the post from Lena/Virtual_Lena.
Parasocial relationship (Lim et al., 2020, 1 = Strongly Disagree, 7 = Strongly Agree)
If Lena/Virtual_Lena would appear live in another post, I would view that post.
I would miss Lena/Virtual_Lena when she was on vacation.
I find Lena/Virtual_Lena's broadcast to be attractive.
I see Lena/Virtual_Lena as a natural, down-to-earth person.
Brand interest (Machleit et al., 1993, 1 = Strongly Disagree, 7 = Strongly Agree)

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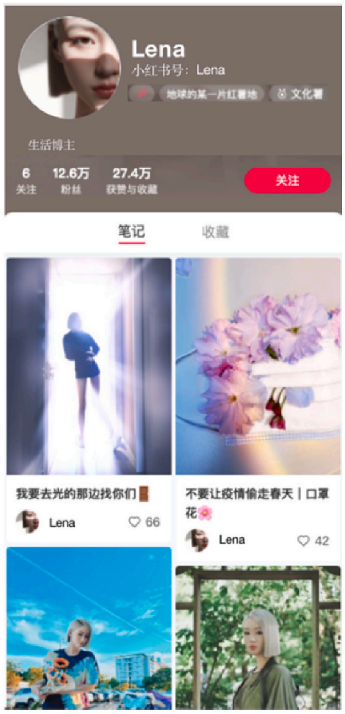
Emotional engagement (Lim et al., 2015, 1 = Strongly Disagree, 7 = Strongly Agree)
I am intrigues by Caltin.
I would like to know more about Caltin.
Learning more about Caltin would be useless.
I am little curious about Caltin.
Purchase intention (Janssen et al., 2022, 1 = Strongly Disagree, 7 = Strongly Agree)
The next time you are looking to purchase this type of product, how likely are you to buy Caltin's products?

Appendix B

Experiment Stimuli

Study 1

Human influencer



Human-like virtual influencer



Brand endorsement



Study 2

Image* Human influencer

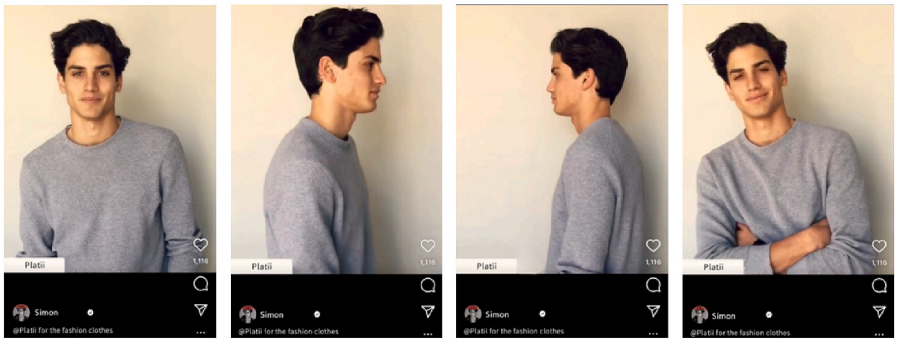
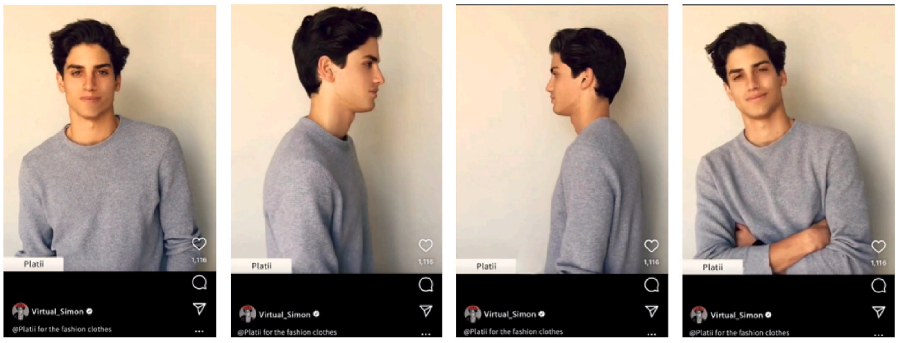
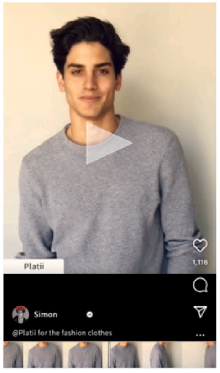


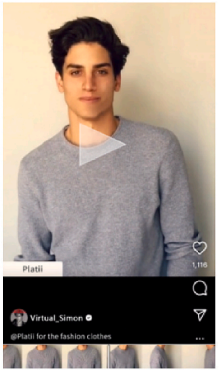
Image* Human-like virtual influencer



Video* Human influencer



Video* Human-like virtual influencer



Appendix C

Construct reliability and validity in Study 1& Study 2

Constructs	Items	Study 1					Study 2				
		Factor Loading	Cronbach's α	rho_A	CR	AVE	Factor Loading	Cronbach's α	rho_A	CR	AVE
Emotional engagement	EE1	0.842	0.823	0.824	0.884	0.657	0.838	0.835	0.836	0.890	0.671
	EE2	0.876					0.856				
	EE3	0.701					0.737				
	EE4	0.814					0.840				
Parasocial relationship	PR1	0.814	0.806	0.810	0.873	0.633	0.741	0.663	0.667	0.799	0.501

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Constructs	Items	Study 1					Study 2				
		Factor Loading	Cronbach's α	rho_A	CR	AVE	Factor Loading	Cronbach's α	rho_A	CR	AVE
	PR2	0.821					0.744				
	PR3	0.829					0.730				
	PR4	0.715					0.604				
Brand interest	BI1	0.845	0.835	0.840	0.890	0.670	0.763	0.776	0.778	0.857	0.599
	BI2	0.836					0.795				
	BI3	0.760					0.730				
	BI4	0.830					0.807				

Appendix D

Table D1
Selected Literature on Comparison of Human and Virtual Influencer

Literature	Focal attributes	Context	Theory	Attitude	Primary Findings
Arsenyan and Mirowska (2021)	Source type (human vs. human-like virtual vs. anime-like virtual)	Interaction	Not discussed	Negative	Compared to humans and anime-like virtual influencers, human-like virtual influencers receive significantly lower positive reactions.
Park et al. (2021)	Source type (virtual vs. human); source characteristics (singles vs. multiple)	Interaction	Not discussed	Negative	Virtual influencers lead to negative feelings of users compared with human.
Thomas and Fowler (2021)	Source type (AI vs. celebrity)	Brand endorsement of transgressions	Not discussed	Positive	AI influencers contribute to positive brand benefits the same like human endorsers.
Franke et al. (2022)	Source type (virtual vs. human)	Advertising	Not discussed	Both	Virtual influencers show lower positive attitudes toward human endorsers in advertising campaigns.
Stein et al. (2022)	Source type (virtual vs. human)	Video streaming	Parasocial interaction theory	Both	Virtual influencers do not lead to significant different in parasocial responses, but lead to lower mental human-likeness and similarity than human influencers.
Deng and Jiang (2023)	Source type (virtual vs. human)	Appearance anxiety	Social comparison theory	Positive	Virtual influencers lead to a lower appearance anxiety than real human.
This article	Source type (virtual vs. human)	Endorsement effectiveness	Parasocial relationship theory	Negative	Virtual influencers lead to a lower endorsement effect than human influencers via consumer's emotional engagement and PSRs.

References

Alboqami, H., 2023. Trust me, I'm an influencer! - causal recipes for customer trust in artificial intelligence influencers in the retail industry. J. Retailing Consum. Serv. 72, 1–12. https://doi.org/10.1016/j.jretconser.2022.103242.

Arsenyan, J., Mirowska, A., 2021. Almost human? A comparative case study on the social media presence of virtual influencers. Int. J. Hum. Comput. Stud. 155, 1–16. https://doi.org/10.1016/j.ijhcs.2021.102694.

Bi, N.C., Zhang, R., 2022. "I will buy what my 'friend'recommends": the effects of parasocial relationships, influencer credibility and self-esteem on purchase intentions. J. Res. Indian Med. 17 (2), 157–175. https://doi.org/10.1108/JRIM-08-2021-0214.

Boerman, S.C., 2020. The effects of the standardized Instagram disclosure for micro-and meso-influencers. Comput. Hum. Behav. 103, 199–207. https://doi.org/10.1016/j.chb.2019.09.015.

Breves, P., Amrehn, J., Heidenreich, A., Liebers, N., Schramm, H., 2021. Blind trust? The importance and interplay of parasocial relationships and advertising disclosures in explaining influencers' persuasive effects on their followers. Int. J. Advert. 40 (7), 1209–1229. https://doi.org/10.1080/02650487.2021.1881237.

Bu, Y., Parkinson, J., Thaichon, P., 2022. Influencer marketing: homophily, customer value co-creation behaviour and purchase intention. J. Retailing Consum. Serv. 66, 1–17. https://doi.org/10.1016/j.jretconser.2021.102904.

Casaló, L.V., Flavián, C., Ibáñez-Sánchez, S., 2020. Influencers on Instagram: antecedents and consequences of opinion leadership. J. Bus. Res. 117, 510–519. https://doi.org/10.1016/j.jbusres.2018.07.005.

Cheung, M.L., Leung, W.K., Aw, E.C.-X., Koay, K.Y., 2022. "I follow what you post!": the role of social media influencers' content characteristics in consumers' online brand-related activities (COBRAs). J. Retailing Consum. Serv. 66, 1–14. https://doi.org/10.1016/j.jretconser.2022.102940.

Chung, S., Cho, H., 2017. Fostering parasocial relationships with celebrities on social media: implications for celebrity endorsement. Psychol. Market. 34 (4), 481–495. https://doi.org/10.1002/mar.21001.

Deng, Z., Benckendorff, P., Wang, J., 2022. From interaction to relationship: rethinking parasocial phenomena in travel live streaming. Tourism Manag. 93, 1–15. https://doi.org/10.1016/j.tourman.2022.104583.

Deng, F., Jiang, X., 2023. Effects of human versus virtual human influencers on the appearance anxiety of social media users. J. Retail. Consum. Serv. 71, 103233.

Derdenger, T.P., Li, H., Srinivasan, K., 2018. Firms' strategic leverage of unplanned exposure and planned advertising: an analysis in the context of celebrity endorsements. J. Market. Res. 55 (1), 14–34. https://doi.org/10.1509/jmr.16.0260.

Dibble, J.L., Hartmann, T., Rosaen, S.F., 2016. Parasocial interaction and parasocial relationship: conceptual clarification and a critical assessment of measures. Hum. Commun. Res. 42 (1), 21–44. https://doi.org/10.1111/hcre.12063.

Djafarova, E., Rushworth, C., 2017. Exploring the credibility of online celebrities' Instagram profiles in influencing the purchase decisions of young female users. Comput. Hum. Behav. 68, 1–7. https://doi.org/10.1016/j.chb.2016.11.009.

Du, Y., Rui, J.R., Yu, N., 2023. How parasocial relationship and influencer-product congruence shape audience's attitude towards product placement in online videos: the mediation role of reactance. Psychol. Res. Behav. Manag. 1315–1329. https://www.tandfonline.com/doi/full/10.2147/PRBM.S406558.

Farivar, S., Wang, F., 2022. Effective influencer marketing: a social identity perspective. J. Retailing Consum. Serv. 67, 1–10. https://doi.org/10.1016/j.jretconser.2022.103026.

Farivar, S., Wang, F., Yuan, Y., 2021. Opinion leadership vs. para-social relationship: key factors in influencer marketing. J. Retailing Consum. Serv. 59, 1–11. https://doi.org/10.1016/j.jretconser.2020.102371.

Feng, Y., Chen, H., Kong, Q., 2021. An expert with whom I can identify: the role of narratives in influencer marketing. Int. J. Advert. 40 (7), 972–993. https://doi.org/10.1080/02650487.2020.1824751.

Franke, C., Groeppel-Klein, A., Müller, K., 2022. Consumers' responses to virtual influencers as advertising endorsers: novel and effective or uncanny and deceiving? J. Advert. 1–17.

Geyser, W., 2022. What Is Influencer Marketing? – the Ultimate Guide for 2022. August 16th. Retrieved from. https://influencermarketinghub.com/influencer-marketing/.

Giles, D.C., 2002. Parasocial interaction: a review of the literature and a model for future research. Media Psychol. 4 (3), 279–305. https://doi.org/10.1207/S1532785XMEP0403_04.

Gray, K., Wegner, D.M., 2012. Feeling robots and human zombies: mind perception and the uncanny valley. Cognition 125 (1), 125–130. https://doi.org/10.1016/j.cognition.2012.06.007.

Guadagno, R.E., Rempala, D.M., Murphy, S., Okdie, B.M., 2013. What makes a video go viral? An analysis of emotional contagion and Internet memes. Comput. Hum. Behav. 29 (6), 2312–2319. https://doi.org/10.1016/j.chb.2013.04.016.

Hartmann, T., Goldhoorn, C., 2011. Horton and Wohl revisited: exploring viewers' experience of parasocial interaction. J. Commun. 61 (6), 1104–1121. https://doi.org/10.1111/j.1460-2466.2011.01595.x.

Hayes, A.F., 2018. Partial, conditional, and moderated moderated mediation: quantification, inference, and interpretation. Commun. Monogr. 85 (1), 4–40. https://doi.org/10.1080/03637751.2017.1352100.
Hill, J., Ford, W.R., Farreras, I.G., 2015. Real conversations with artificial intelligence: a comparison between human–human online conversations and human–chatbot conversations. Comput. Hum. Behav. 49, 245–250. https://doi.org/10.1016/j.chb.2015.02.026.
Hilvert-Bruce, Z., Neill, J.T., Sjöblom, M., Hamari, J., 2018. Social motivations of live-streaming viewer engagement on Twitch. Comput. Hum. Behav. 84, 58–67. https://doi.org/10.1016/j.chb.2018.02.013.
Horton, D., Richard Wohl, R., 1956. Mass communication and para-social interaction: observations on intimacy at a distance. Psychiatry 19 (3), 215–229. https://doi.org/10.1080/00332747.1956.11023049.
Hu, L., Min, Q., Han, S., Liu, Z., 2020. Understanding followers' stickiness to digital influencers: the effect of psychological responses. Int. J. Inf. Manag. 54, 1–14. https://doi.org/10.1016/j.ijinfomgt.2020.102169.
Huang, A.H., Yen, D.C., Zhang, X., 2008. Exploring the potential effects of emoticons. Inf. Manag. 45 (7), 466–473. https://doi.org/10.1016/j.im.2008.07.001.
Hwang, K., Zhang, Q., 2018. Influence of parasocial relationship between digital celebrities and their followers on followers' purchase and electronic word-of-mouth intentions, and persuasion knowledge. Comput. Hum. Behav. 87, 155–173. https://doi.org/10.1016/j.chb.2018.05.029.
Janssen, L., Schouten, A.P., Croes, E.A., 2022. Influencer advertising on Instagram: product-influencer fit and number of followers affect advertising outcomes and influencer evaluations via credibility and identification. Int. J. Advert. 41 (1), 101–127. https://doi.org/10.1080/02650487.2021.1994205.
Ki, C.-W.C., Cuevas, L.M., Chong, S.M., Lim, H., 2020. Influencer marketing: social media influencers as human brands attaching to followers and yielding positive marketing results by fulfilling needs. J. Retailing Consum. Serv. 55, 1–11. https://doi.org/10.1016/j.jretconser.2020.102133.
Kim, H., 2022. Keeping up with influencers: exploring the impact of social presence and parasocial interactions on Instagram. Int. J. Advert. 41 (3), 414–434. https://doi.org/10.1080/02650487.2021.1886477.
Kim, M., Kim, J., 2020. How does a celebrity make fans happy? Interaction between celebrities and fans in the social media context. Comput. Hum. Behav. 111, 1–11. https://doi.org/10.1016/j.chb.2020.106419.
Kim, H., Park, M., 2023. Virtual influencers' attractiveness effect on purchase intention: a moderated mediation model of the Product–Endorser fit with the brand. Comput. Hum. Behav. 143, 1–10. https://doi.org/10.1016/j.chb.2023.107703.
Kowert, R., Daniel Jr., E., 2021. The one-and-a-half sided parasocial relationship: the curious case of live streaming. Computers in Human Behavior Reports 4, 1–7. https://doi.org/10.1016/j.chbr.2021.100150.
Labrecque, L.I., 2014. Fostering consumer–brand relationships in social media environments: the role of parasocial interaction. J. Interact. Market. 28 (2), 134–148. https://doi.org/10.1016/j.intmar.2013.12.003.
Lee, M., Hong, J.H., Chung, S., Back, K.-J., 2021. Exploring the roles of DMO's social media efforts and information richness on customer engagement: empirical analysis on Facebook event pages. J. Trav. Res. 60 (3), 670–686. https://doi.org/10.1177/0047287520934874.
Lefebvre, S., Cowart, K., 2021. An investigation of influencer body enhancement and brand endorsement. J. Serv. Market. 36 (1), 59–72. https://doi.org/10.1108/JSM-12-2020-0509.
Li, B., Chen, S., Zhou, Q., 2023. Empathy with influencers? The impact of the sensory advertising experience on user behavioral responses. J. Retailing Consum. Serv. 72, 1–12. https://doi.org/10.1016/j.jretconser.2023.103286.
Li, J., Huang, J., Li, Y., 2023. Examining the effects of authenticity fit and association fit: a digital human avatar endorsement model. J. Retailing Consum. Serv. 71, 1–11. https://doi.org/10.1016/j.jretconser.2022.103230.
Liljander, V., Gummerus, J., Söderlund, M., 2015. Young consumers' responses to suspected covert and overt blog marketing. Internet Res. 25 (4), 610–632. https://doi.org/10.1108/IntR-02-2014-0041.
Lim, J.S., Choe, M.-J., Zhang, J., Noh, G.-Y., 2020. The role of wishful identification, emotional engagement, and parasocial relationships in repeated viewing of live-streaming games: a social cognitive theory perspective. Comput. Hum. Behav. 108, 1–10. https://doi.org/10.1016/j.chb.2020.106327.
Lim, J.S., Hwang, Y., Kim, S., Biocca, F.A., 2015. How social media engagement leads to sports channel loyalty: mediating roles of social presence and channel commitment. Comput. Hum. Behav. 46, 158–167. https://doi.org/10.1016/j.chb.2015.01.013.
Longoni, C., Cian, L., 2022. Artificial intelligence in utilitarian vs. hedonic contexts: the "word-of-machine" effect. J. Market. 86 (1), 91–108. https://doi.org/10.1177/0022242920957347.
Lou, C., Kiew, S.T.J., Chen, T., Lee, T.Y.M., Ong, J.E.C., Phua, Z., 2022. Authentically fake? How consumers respond to the influence of virtual influencers. J. Advert. 1–18. https://doi.org/10.1080/00913367.2022.2149641.
Lou, C., Yuan, S., 2019. Influencer marketing: how message value and credibility affect consumer trust of branded content on social media. J. Interact. Advert. 19 (1), 58–73. https://doi.org/10.1080/15252019.2018.1533501.
Luo, X., Tong, S., Fang, Z., Qu, Z., 2019. Frontiers: machines vs. humans: the impact of artificial intelligence chatbot disclosure on customer purchases. Market. Sci. 38 (6), 937–947. https://doi.org/10.1287/mksc.2019.1192.
Machleit, K.A., Allen, C.T., Madden, T.J., 1993. The mature brand and brand interest: an alternative consequence of ad-evoked affect. J. Market. 57 (4), 72–82. https://doi.org/10.1177/002224299305700406.
Masuda, H., Han, S.H., Lee, J., 2022. Impacts of influencer attributes on purchase intentions in social media influencer marketing: mediating roles of characterizations. Technol. Forecast. Soc. Change 174, 1–12. https://doi.org/10.1016/j.techfore.2021.121246.
Miao, F., Kozlenkova, I.V., Wang, H., Xie, T., Palmatier, R.W., 2022. An emerging theory of avatar marketing. J. Market. 86 (1), 67–90. https://doi.org/10.1177/0022242921996646.
Molenaar, K., 2022. *Discover the Top 15 Virtual Influencers for 2022 – Listed and Ranked!* Influencer Marketing Hub. Retrieved from. https://influencermarketinghub.com/virtual-influencers/#toc-1.
Molinillo, S., Aguilar-Illescas, R., Anaya-Sánchez, R., Vallespín-Arán, M., 2018. Exploring the impacts of interactions, social presence and emotional engagement on active collaborative learning in a social web-based environment. Comput. Educ. 123, 41–52. https://doi.org/10.1016/j.compedu.2018.04.012.
Mou, Y., Xu, K., 2017. The media inequality: comparing the initial human-human and human-AI social interactions. Comput. Hum. Behav. 72, 432–440. https://doi.org/10.1016/j.chb.2017.02.067.
Mozafari, N., Weiger, W.H., Hammerschmidt, M., 2021. Trust me, I'm a bot–repercussions of chatbot disclosure in different service frontline settings. J. Serv. Manag. 33 (2), 221–245. https://doi.org/10.1108/JOSM-10-2020-0380.
Munnukka, J., Maity, D., Reinikainen, H., Luoma-aho, V., 2019. "Thanks for watching". The effectiveness of YouTube vlogendorsements. Comput. Hum. Behav. 93, 226–234. https://doi.org/10.1016/j.chb.2018.12.014.
Park, G., Nan, D., Park, E., Kim, K.J., Han, J., Del Pobil, A.P., 2021 January. Computers as social actors? Examining how users perceive and interact with virtual influencers on social media. In: 2021 15th International Conference on Ubiquitous Information Management and Communication (IMCOM). IEEE, pp. 1–6.
Perez-Vega, R., Kaartemo, V., Lages, C.R., Razavi, N.B., Männistö, J., 2021. Reshaping the contexts of online customer engagement behavior via artificial intelligence: a conceptual framework. J. Bus. Res. 129, 902–910. https://doi.org/10.1016/j.jbusres.2020.11.002.
Ramasubramanian, S., Kornfield, S., 2012. Japanese anime heroines as role models for US youth: wishful identification, parasocial interaction, and intercultural entertainment effects. Journal of International and Intercultural Communication 5 (3), 189–207. https://doi.org/10.1080/17513057.2012.679291.
Ramirez, J., Artemio, Burgoon, J.K., 2004. The effect of interactivity on initial interactions: the influence of information valence and modality and information richness on computer-mediated interaction. Commun. Monogr. 71 (4), 422–447. https://doi.org/10.1080/0363452042000307461.
Reinikainen, H., Munnukka, J., Maity, D., Luoma-Aho, V., 2020. 'You really are a great big sister'–parasocial relationships, credibility, and the moderating role of audience comments in influencer marketing. J. Market. Manag. 36 (3–4), 279–298. https://doi.org/10.1080/0267257X.2019.1708781.
Rubin, A.M., Perse, E.M., Powell, R.A., 1985. Loneliness, parasocial interaction, and local television news viewing. Hum. Commun. Res. 12 (2), 155–180. https://doi.org/10.1111/j.1468-2958.1985.tb00071.x.
Sands, S., Ferraro, C., Demsar, V., Chandler, G., 2022. False idols: unpacking the opportunities and challenges of falsity in the context of virtual influencers. Bus. Horiz. 65 (6), 777–788. https://doi.org/10.1016/j.bushor.2022.08.002.
Schouten, A.P., Janssen, L., Verspaget, M., 2020. Celebrity vs. Influencer endorsements in advertising: the role of identification, credibility, and Product-Endorser fit. Int. J. Advert. 39 (2), 258–281. https://doi.org/10.1080/02650487.2019.1634898.
Shin, D.-H., 2016. Do users experience real sociability through social TV? Analyzing parasocial behavior in relation to social TV. J. Broadcast. Electron. Media 60 (1), 140–159. https://doi.org/10.1080/08838151.2015.1127247.
Sokolova, K., Kefi, H., 2020. Instagram and YouTube bloggers promote it, why should I buy? How credibility and parasocial interaction influence purchase intentions. J. Retailing Consum. Serv. 53, 1–9. https://doi.org/10.1016/j.jretconser.2019.01.011.
Stein, J.-P., Linda Breves, P., Anders, N., 2022. Parasocial Interactions with Real and Virtual Influencers: the Role of Perceived Similarity and Human-Likeness. New Media & Society, pp. 1–21. https://doi.org/10.1177/14614448221102900.
Stern, B.B., Russell, C.A., Russell, D.W., 2007. Hidden persuasions in soap operas: damaged heroines and negative consumer effects. Int. J. Advert. 26 (1), 9–36. https://doi.org/10.1080/02650487.2007.11072994.
Stubb, C., Colliander, J., 2019. "This is not sponsored content"–The effects of impartiality disclosure and e-commerce landing pages on consumer responses to social media influencer posts. Comput. Hum. Behav. 98, 210–222. https://doi.org/10.1016/j.chb.2019.04.024.
Thomas, V.L., Fowler, K., 2021. Close encounters of the AI kind: use of AI influencers as Brand endorsers. J. Advert. 50 (1), 11–25. https://doi.org/10.1080/00913367.2020.1810595.
Tukachinsky, R., Stever, G., 2019. Theorizing development of parasocial engagement. Commun. Theor. 29 (3), 297–318. https://doi.org/10.1093/ct/qty032.
Tukachinsky, R., Walter, N., Saucier, C.J., 2020. Antecedents and effects of parasocial relationships: a meta-analysis. J. Commun. 70 (6), 868–894. https://doi.org/10.1093/joc/jqaa034.
Vazquez, D., Wu, X., Nguyen, B., Kent, A., Gutierrez, A., Chen, T., 2020. Investigating narrative involvement, parasocial interactions, and impulse buying behaviours within a second screen social commerce context. Int. J. Inf. Manag. 53, 102135 https://doi.org/10.1016/j.ijinfomgt.2020.102135.
Whang, C., Im, H., 2021. "I like Your Suggestion!" the role of humanlikeness and parasocial relationship on the website versus voice shopper's perception of recommendations. Psychol. Market. 38 (4), 581–595. https://doi.org/10.1002/mar.21437.

Yoo, Y., Alavi, M., 2001. Media and group cohesion: relative influences on social presence, task participation, and group consensus. MIS Q. 371–390. https://doi.org/10.2307/3250922.

Youn, S., Jin, S.V., 2021. In AI we trust?" The effects of parasocial interaction and technopian versus luddite ideological views on chatbot-based customer relationship management in the emerging "feeling economy. Comput. Hum. Behav. 119, 106721 https://doi.org/10.1016/j.chb.2021.106721.

Yuan, S., Lou, C., 2020. How social media influencers foster relationships with followers: the roles of source credibility and fairness in parasocial relationship and product interest. J. Interact. Advert. 20 (2), 133–147. https://doi.org/10.1080/15252019.2020.1769514.